

Crowning the Metropolis: Skylines, Land Values, and Urban Population

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Abstract

The height of American cities' tallest buildings correlates strongly with their greater metropolitan area's population. This relationship is explained in a monocentric city model through central land values, which rise proportionally with population, income, and land unavailability. Higher values incentivize developers to substitute capital for land when producing floor space, raising building heights. Using a recursive simultaneous equation system, we endogenize income through agglomeration economies, and population through Zipf's Law, while accounting for construction costs, geology, and land-use regulations. Testing these economic relationships we find this analytically tractable model neatly ties together metropolitan population, central land values, and skyline heights.

1 Introduction

A curious fact about U.S. cities is that the heights of their tallest buildings predict the populations of their metropolitan areas rather well. The correlation coefficient between population and the average height of the tallest ten buildings – both expressed in logarithms, and weighing by metro population – is 0.88. The regression line, seen in Figure 1 panel A, finds that a 10 percent higher population predicts almost 4 percent taller buildings across most of the sample. This reflects that average skyscraper heights vary in range by a factor of 10 — from one hundred to one thousand feet — while metro populations vary by a factor 400 — from 50 thousand to 20 million heads. Conversely, by observing the height of a city’s skyline, one can predict how many residents are within commuting distance with decent accuracy.

No policy or law — neither legal nor natural — ordains a close connection between metro population and building heights. Instead, this paper proposes the two quantities are connected by a key price — the value of land in central cities — through two simple structural models. The first is the canonical monocentric city model of [Alonso \(1964\)](#), [Mills \(1967\)](#), [Muth \(1969\)](#), which explains how larger populations raise central land values. The second models how higher central land values raise building heights, as developers cost-minimize by substituting capital for land ([Muth, 1969](#); [Ahlfeldt and McMillen, 2018](#)). Work by [Albouy et al. \(2018\)](#) provides estimates of these central land values for U.S. metropolitan areas, allowing us to jointly estimate and test these two models empirically.¹

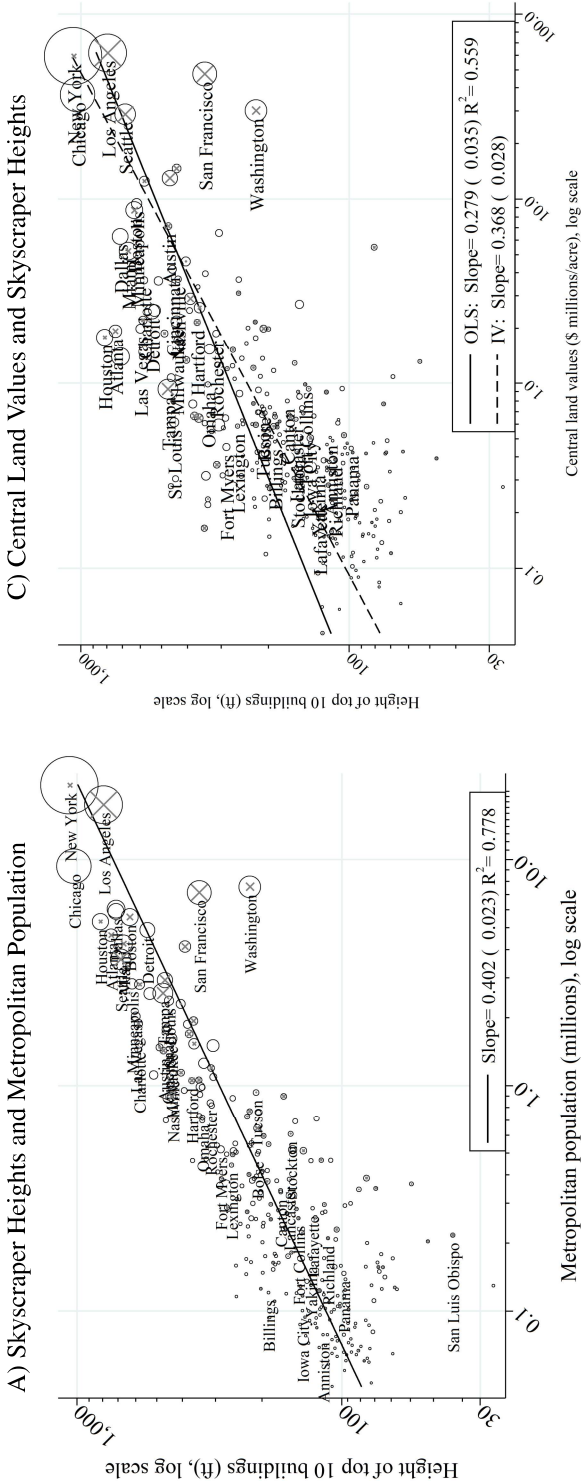
The cross-metro relationship between population and central land values, shown in figure 1 panel B, tests a hypothesis foundational in urban economics. Here, the correlation coefficient is 0.89 and the regression coefficient is almost one. This exact proportionality of central land values and metro populations is predicted by a two-dimensional monocentric-city model under standard conditions ([Combes et al., 2012](#); [Duranton and Puga, 2015](#)). Deviations from the regression line are explained by income and how cities’ shapes provide access to develop-able land, echoing themes in [Saiz \(2010\)](#) and [Harari \(2020a\)](#). In all, it is both surprising and satisfying that a simple yet precise prediction about land values fits the data so well.²

Figure 1 panel C, is inspired by the economic model of how high land values push developers to build taller to minimize costs. The slope of the line reflects how much these developers substitute capital for land in creating floor space. While other analyses have considered this relationship

¹Previous land-value estimates in the literature, e.g. [Davis and Palumbo \(2008\)](#), have focused on residential land under single-family buildings, which is typically peripheral, and thus not appropriate for central multi-story buildings.

²Using data for French municipalities, [Combes et al. \(2018\)](#) estimate an elasticity of land values with respect to population near 0.7. They define central land values differently from us, relying more on distance controls.

Figure 1: Skyscraper Heights, Metropolitan Population, and Central Land Values



Notes: Robust standard errors given in parentheses. Metropolitan areas refer to 2010 metropolitan statistical areas (MSAs), consolidated, when possible. Heights from SkyscraperPage.com and Emporis are for 2010. Regressions are weighted by the inverse of the population rank. The size of the marker is proportional to the metro population times the arc of expansion, as explained in the text. “IV” refers to the use of metropolitan population as an instrumental variable for land values, as explained in the text.

within a city — e.g. [Ahlfeldt and McMillen \(2018\)](#) examine variation within Chicago; [Ahlfeldt and Barr \(2022\)](#) within New York — ours applies across cities. Besides showing that these relationship apply broadly, the first structural model suggests a possible instrumental variable for these land values, namely metropolitan population. Furthermore, the cross-metro design allows us to consider the influence of variation in construction costs and regulatory rigidity.³

To complete a full empirical system of equations, we supplement three ancillary structural relationships to the two described above. The first two allow incomes and construction costs to be endogenous to metropolitan population and city shape. Specifically, the first relates incomes to metro population, a relationship which has its own attendant literature, e.g. [Glaeser and Gottlieb \(2008\)](#), [Melo et al. \(2009\)](#), and [Albouy \(2016\)](#). The second relates how these incomes, tied to wages, are reflected in construction costs, as touched on briefly in [Albouy and Ehrlich \(2018\)](#). The third, and last, relationship is the well-known Zipf’s Law of Cities ([Gabaix, 1999](#)), which relates how a metro’s population is inversely proportional to its rank. Population rank may be used to instrument population differences that may be due to transitory or measurement issues ([Ahlfeldt and Pietrostefani, 2019](#)).

2 Structural Equations

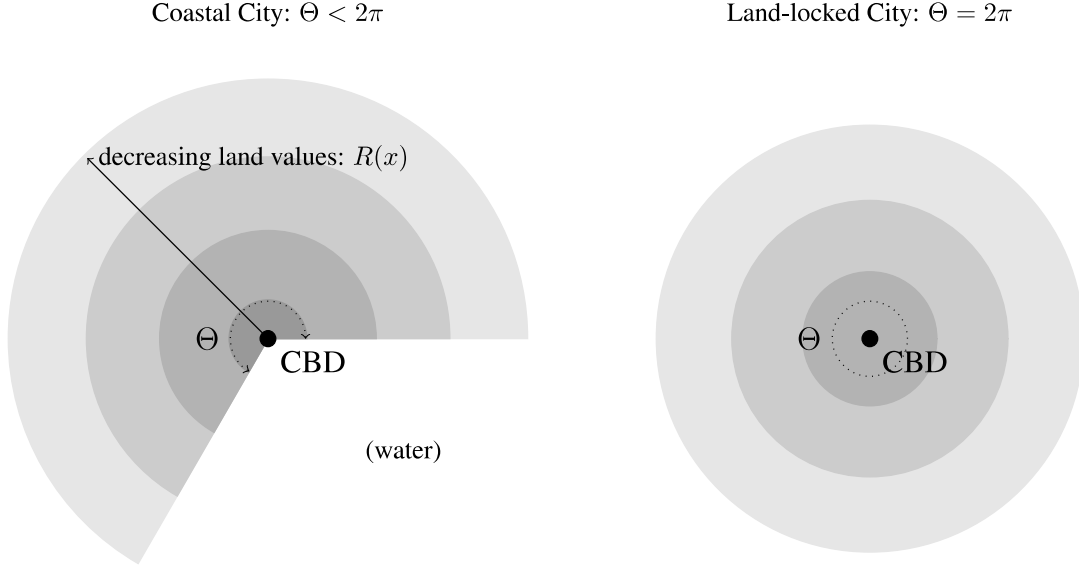
In our model, each city (i.e. metro area) j expands from a central business district (CBD), where skyscrapers are built. For simplicity, land values are uniform within the CBD. In theory, the CBD is a point; in practice, it is larger but much smaller than outlying areas. Moving out from the CBD, land is available on a potentially endless circular sector. This sector has a fixed (exogenous) arc of expansion, Θ^j , measured in radians, as illustrated in Figure 2. An arc of 2π corresponds to a landlocked city; $\Theta^j < 2\pi$, describes a city with a coastal line. In the base model, we assume the homogenous metropolitan population N^j and arc of expansion Θ^j are exogenous.

The structural relationships explain each city’s income, M^j ; construction input prices, V^j ; central land values, $R^j(0)$;⁴ and building heights, H^j . Together, the structural equations combine recursively to create a reduced-form system of equations. To follow the trail of causality intuitively, we present the system in reverse order of their recursion. The Appendix offers greater detail and derivations. We also use lower-case letters to denote logarithms, i.e. $x^j \equiv \ln X^j$ for any X .

³[Brueckner and Singh \(2020\)](#) estimate the local elasticity of land values to floor-to-area ratios in 5 U.S. cities to infer the stringency of regulations. Other work by Barr and Jedwab consider how skyscraper heights vary with income.

⁴Land values $R^j(x)$, change as a function of distance from the city center, x .

Figure 2: Monocentric Cities with Equal Population but Different Arcs of Expansion



2.1 Skyscraper Heights

Skyscraper developers minimize costs in producing floor space, taking as given the price of land in the CBD, $R^j(0)$, and construction inputs with price, V^j . This means developers produce skyscrapers with an ideal height of

$$h^j = h^0 + \sigma^* \cdot (r^j(0) - v^j + b^j), \quad (1)$$

The $*$ on σ signifies that it is a transformation of the elasticity of substitution between capital and land. That elasticity is smaller when more land is required around the building's footprint as it gets taller; it is larger to the degree that capital intensity rises on the margin as the building gets taller, as greater heights reveal even greater capital use.⁵ $b^j \equiv \ln B^j$ captures differences in factor biases – favoring capital over land – that may differ across cities.

⁵As shown in the appendix $\sigma^* = \sigma(1 + \zeta - \xi)^{-1}$, where ζ is the elasticity of capital to floor-space with respect to building height, and ξ is the elasticity of the land relative to the footprint of the actual building. Using data on a sub-sample of buildings from Chicago, [Ahlfeldt and McMillen \(2018\)](#) estimate these adjustments increase the estimated elasticity of substitution by roughly 30 percent.

2.2 Central Land Values

The CBD abuts land used for residential purposes. Housing producers generate housing services from land and construction inputs with Cobb-Douglas technology and constant returns to scale. This implies a fixed cost share of land. Residents consume housing and non-housing goods according to Cobb-Douglas preferences. This creates an iso-elastic demand for housing and keeps the proportion of income spent on housing fixed. They also pay commuting costs that increase with distance, x from the CBD. These costs enter multiplicatively into the utility function with a negative exponent. These assumptions imply land values, $R^j(x)$, fall with distance x from the CBD to compensate households for commuting costs, following a power law.

The Cobb-Douglas economy implies that land values are proportional to mean income and population of the city. Multiplicative commuting costs keep this proportionality even as values fall from the center (Combes et al., 2012). Furthermore, land values vary inversely with the arc of expansion. This is seen in figure 2, where the coastal city on the left has higher central land values, as well as densities, while having the same population as the land-locked city on the right. The model also includes an exogenous shifter due to commuting costs, F^j , which can arise from (unmodeled) infrastructure or topography:

$$r^j(0) = r^0(0) + n^j + m^j - \theta^j - f^j. \quad (2)$$

Differences in central land values are driven entirely by differential commuting costs, even though the variation described in equation (2) does not depend on parameters describing commuting costs or boundaries of the city.⁶

2.3 Construction Input Prices

Construction inputs for skyscrapers consist of material and labor used in installation. Material inputs are provided in competitive markets and are shipped without cost, meaning their unit price is uniform across cities. Labor must be hired in the local market given the prevailing wage — assumed uniform within a city — which in turn determines local incomes. Therefore, construction input prices are related to city incomes through a cost function given by:

$$v^j = v^0 + a \cdot m^j - a_v^j, \quad (3)$$

⁶We provide a proof in the appendix.

Technically, the parameter a is the cost share of labor divided by labor's share of household income. a_v^j captures any potential productivity shifters in construction inputs across cities.⁷

2.4 Local Incomes

Urban economists have long described how agglomeration economies cause incomes to rise — for a meta-analysis see [Melo et al. \(2009\)](#). Sidestepping any subtleties, we model the relationship between income and population with the standard reduced-form power function:

$$m^j = m^0 + \gamma \cdot n^j + a_m^j \quad (4)$$

Here γ is the agglomeration productivity parameter, and $a_m^j \equiv \ln A_m^j$ accounts for differences in city productivity unaccounted for by this simple agglomeration model.

In this economic model, what matters is how income affects the demand for land. Thus, it is not important here for γ to provide a pure estimate of the effect of population alone. Larger cities may attract more skilled workers, or cause individuals to work more, making γ larger than it would be if skills levels and effort were held constant.

2.5 Zipf's Law of Cities

The population of a city is clearly endogenous to its local productivity and quality of life ([Albouy and Stuart, 2020](#)), yet the land model implicitly allows for this simultaneity ([Combes et al., 2012](#)). In an open monocentric system of cities — see [Arnott and Stiglitz \(1979\)](#) and [Hoehn et al. \(1987\)](#) — local amenities raise land values precisely by attracting population, who bid up the price of land where commuting costs are low. Thus, population may be taken as exogenous to land values in so far as they are raised through this bidding behavior, including endogenous agglomeration effects on income. Similarly, population is exogenous to skyscraper heights so long as heights are raised only in so far as greater population raises land and construction input costs.

We do consider a possible instrument for metropolitan population inspired by [Ahlfeldt and Pietrostefani \(2019\)](#). Namely, we use population rank as an instrument for population:

$$n^j = n^0 + p^j + e_p^j \quad (5)$$

where $p^j \equiv \ln P^j$ is the inverse of the metro area's position in the population ranking and e_p^j is

⁷Note Cobb-Douglas technology provides an exact first-order approximation in logarithms for any cost function, as in (3), but not for quantity-price relationships as in (1).

an error term. This instrument may help reduce biases if endogenous forces change population, but not enough to change population rank. The population rank instrument not only reduces the potential biased cause by endogeneity but also addresses potential measurement errors that can affect the population count but are unlikely to be large enough to modify the ranking. This may help if population rank is better measured than population. It may also help if endogeneity problems are. Rank may serve as a better idealization, if say, the metro population of Los Angeles isn't correctly measured, or if unobserved factors affect the population measurement but not the rank. In this vein, it is worth addressing the potential endogeneity of population—population can affect building heights, land values, construction cost, and income through mechanisms that are not explicit in the model. To do this we build an instrumental variable based on the Zipf's power law of the ranking of cities listed by population:

3 Econometric Model

The recursive system of structural equations in (1), (2), (3), and (4), is easily transformed into a reduced-form system of four equations with two exogenous variables — population and arc of expansion. Adding control variables, X^j , that include a constant this system may be written as

$$h^j = \pi_{hn}n^j + \pi_{h\theta}\theta^j + X^j\beta_h + \varepsilon_h^j \quad (6a)$$

$$r^j(0) = \pi_{rn}n^j + \pi_{r\theta}\theta^j + X^j\beta_r + \varepsilon_r^j \quad (6b)$$

$$v^j = \pi_{vn}n^j + \pi_{v\theta}\theta^j + X^j\beta_v + \varepsilon_v^j \quad (6c)$$

$$m^j = \pi_{mn}n^j + \pi_{m\theta}\theta^j + X^j\beta_m + \varepsilon_m^j \quad (6d)$$

where the eight reduced form parameters in (6) are restricted to depend on the three structural parameters γ , a , and σ^* .

$$\pi_{hn} = \sigma^*(1 + \gamma - a\gamma) \quad \pi_{h\theta} = -\sigma^* \quad (7a)$$

$$\pi_{rn} = 1 + \gamma \quad \pi_{r\theta} = -1 \quad (7b)$$

$$\pi_{vn} = a\gamma \quad \pi_{v\theta} = 0 \quad (7c)$$

$$\pi_{mn} = \gamma \quad \pi_{m\theta} = 0 \quad (7d)$$

Reading the system upwards relays the recursive structure of the model. The model is over-identified as it contains five restrictions. We organize these restrictions to be independent to test the economic

hypotheses.

- I $\pi_{hn}(\pi_{r\theta} - \pi_{v\theta}) = \pi_{h\theta}(\pi_{rn} - \pi_{vn})$. This is equivalent to the restriction that each instrument produces the same elasticity of substitution. $\sigma^* = \frac{\pi_{hn}}{\pi_{rn} - \pi_{vn}} = \frac{\pi_{h\theta}}{\pi_{r\theta} - \pi_{v\theta}}$.
- II $\pi_{rn} = 1 + \pi_{mn}$ This means central land values rise proportionally with population and income.
- III $\pi_{r\theta} = -1 + \pi_{m\theta}$. This means central land values fall proportionally with the arc of expansion.
- IV $\pi_{vn}\pi_{m\theta} = \pi_{v\theta}\pi_{mn}$. This is equivalent to the restriction that each instrument produces the same cost parameter $a = \frac{\pi_{vn}}{\pi_{mn}} = \frac{\pi_{v\theta}}{\pi_{m\theta}}$.
- V $\pi_{m\Theta} = 0$ There is no relationship between the arc of expansion and income.

Multiple reasons may cause over-identification tests to fail. We consider possible violations in reverse order.

Restriction V may fail This latter condition arguably adds a check that there are no omitted variables correlated with the arc of expansion that are also related to income. For example, cities with a smaller arc could have lower trade costs with other cities or countries, or attract a more skilled population.

Restriction IV, regarding construction costs, is easier to understand in conjunction with restriction V. This means that $\pi_{v\theta} = 0$, so that differences in construction input prices are due to local wages alone. This may fail if geographically constrained cities suffer additional costs of shipping building materials into the CBD.⁸

Restrictions II and III, regarding central land values, could fail for a number of reasons. The proportionality of these land values to population and income does not require a Cobb-Douglas economy. According to [Duranton and Puga \(2015\)](#), the key assumptions are “isoelastic demand for housing, constant returns in production, and multiplicatively separable commuting cost that directly enter the utility function.”⁹ Central land values could also fail to fall proportionally with the arc of expansion, if for example, such cities invested more heavily in infrastructure.¹⁰

Restriction I, regarding building heights, may fail if developers value height for advertising or other symbolic reasons, they may be induced to build taller than cost minimization alone would

⁸Without restriction V, it would imply that the parameter a estimated using Θ as an instrument with indirect least squares is statistically indistinguishable from a estimated using N . Since estimator using Θ In addition, if labor share of construction input cost is lower than labor’s share of income then $a \leq 1$, and thus the inequality $\pi_{vn} < \pi_{mn}$ should hold. It does in all of the specifications we consider below.

⁹Evidence from [Albouy and Ehrlich \(2018\)](#) and [Combes et al. \(2018\)](#) implies that housing demand elasticities are fairly close to one.

¹⁰Restriction III does allow for an effect of arc on income, which restriction V rules out.

dictate. A greater local population may make such advertising more valuable. Similarly, a city with a high arc of expansion may have a skyline more visible to a population more tightly clustered around it. Alternatively, building occupants may value views from higher floors in cities with greater population or with more or less of an arc of expansion.

In lieu of these possible violations, we consider several versions of the model. In the "Full" model all 5 restrictions hold; in the "Main" model, restriction IV may fail; in the "Core" model, only restrictions I, II, and III need to hold.

Adding Zipf's Law from (5) into the system of equation changes the reduced form to depend on inverse population rank, P , instead of population, N . This involves replacing each "n" with "p" in (6d) and adding the reduced-form equation

$$n^j = \pi_{np}p^j + \pi_{n\theta}\theta^j + X^j\beta_n + \varepsilon_n^j \quad (8)$$

Under the Zipf's Law restrictions that $\pi_{np} = 1$ and $\pi_{n\theta} = 0$, restrictions I through V are essentially unchanged. We consider a more general model where these may fail to hold independently of the other restrictions. The Appendix explains this in full.

We incorporate several control variables into the model that may affect land values or skyscraper height. Land-use restrictions in the city as a whole should reduce density, reducing commuting efficiency to peripheral land. For a given population this raises the value of land in the center, $r^j(0)$, mathematically lowering f^j in (2). Land-use restrictions should also obviously reduce building height. Formally, this creates a factor bias limiting capital's productivity, lowering b^j in (1).

We also include a control for building age. Older buildings may be shorter due to expensive building technologies lowering b^j . More generally, older buildings may be mismatched with current values dynamically, due to their durability. As buildings age and land appreciates, they may not fully reflect the value of the land underneath them. This motivates not only a control but a robustness check, seen in the Appendix, that uses only newer buildings.

Lastly, we control for distance from the surface to bedrock in the CBD. Having to dig deeper to build is thought to increase construction costs and reduce building height (Ahlfeldt and Barr, 2022). It may also have effects on residential density around the CBD.

4 Data

To get overlapping data across all domains, we restrict the analysis to cover the cross section of U.S. metropolitan areas in 2010. The data is organized into 257 metropolitan areas for which we

have data on the main variables from all sources.¹¹ Metropolitan areas refer to 2010 consolidated metropolitan statistical areas (CMSAs) using 1999 OMB definitions. Population numbers and mean family income are taken from the 2010 Census and the 2008-2012 American Community Survey 5 year sample.

Building heights are sourced from SkyscraperPage.com. The website provides data on the tallest high-rises in each U.S. city, including information such as coordinates, height, number of floors, and year built, among other characteristics. Because SkyscraperPage.com did not offer complete data for some of the smaller cities in the sample, we supplemented the dataset using a similar source, Emporis.com. For skyscraper height, we take the average height of the top ten tallest buildings in each metropolitan area. We consider alternative groupings getting similar results, as shown in the appendix Table ???. The question of the number of buildings is driven by data limitations; increasing the number of buildings beyond ten reduces the number of metros in the sample, as some small cities only have a few tall buildings. As we do not have land values across cities for more than a few years, we also collect building ages as additional explanatory variable.

For central land values we follow the same methodology reported in Albouy et al. (2018) with a slight amendment. While this article modeled the city center as being city hall, here we use an amended center around the geographic median of the tallest ten buildings. We then consider prices within a 1 mile radius central land values.

As we do not have a way of systematizing restrictions on building heights, we instead proxy for them using the Wharton Residential Land Use Regulatory Index, using only the index number for the central city, where almost all the skyscrapers are located.¹²

Data on each CMSAs' arc of expansion was first obtained from Stephen Malpezzi. We expanded the data set using Google maps and our own calculations. Construction costs are provided by R.S. Means and cover both materials and installation costs.

5 Results

We present the reduced form specifications from the system in (6) in Table 1, and in Table 2 adding in Zipf's Law from (8). Panel A in each tables presents the unrestricted reduced-form estimates, while Panel B presents tests of the structural restrictions. Columns 1 through 4 present estimates without controls, while columns 5 through 8 use the three controls mentioned earlier.

¹¹Some controls and variables used in alternative specifications, described in the appendix, are only available for a subset of the sample.

¹²While the response rate to this survey was low for all municipalities, it is much higher for central cities.

Table 1: Determinants of Skyscraper Heights, Land Values, Construction Input Prices, and Income:
Reduced Form Regression Results and Structural Restrictions Tests

Dependent Variables	Skyscr. height h^j (1)	Land value r^j (2)	Const. input v^j (3)	H-hold income m^j (4)	Skyscr. height h^j (5)	Land value r^j (6)	Const. input v^j (7)	H-hold income m^j (8)
<i>Panel A: Reduced-Form Parameter Estimates^a</i>								
Log population	0.385 (0.036)	1.002 (0.071)	0.046 (0.010)	0.088 (0.009)	0.380 (0.035)	1.005 (0.073)	0.048 (0.011)	0.084 (0.010)
Arc of expansion (log radians, θ^j)	-0.186 (0.207)	-0.966 (0.491)	-0.094 (0.049)	0.001 (0.045)	-0.118 (0.133)	-1.121 (0.507)	-0.106 (0.049)	-0.002 (0.037)
Regulatory Index (Wharton)					-0.140 (0.036)	0.198 (0.102)	-0.004 (0.013)	0.031 (0.017)
Building Age, years					-0.405 (0.159)	0.477 (0.302)	0.007 (0.034)	0.023 (0.041)
Distance to Bedrock (cms)					-0.037 (0.068)	0.200 (0.166)	0.038 (0.022)	0.052 (0.029)
R^2	0.783	0.816	0.482	0.689	0.835	0.822	0.503	0.706
<i>Panel B: Tests of Structural Restrictions</i>								
I: $\pi_{ln}(\pi_{r\theta} - \pi_{v\theta}) = \pi_{H\theta}(\pi_{rn} - \pi_{vn})$	0.648	Full Model (I-V) 0.0103			0.2918	Full Model(I-V) 0.0070		
II: $\pi_{rn} = 1 + \pi_{mn}$	0.2086	Main Model (I-III & V) 0.3676			0.2451	Main Model (I-III & V) 0.4601		
III: $\pi_{r\theta} = -1 + \pi_{m\theta}$	0.9431	Core Model (I-III) 0.2323			0.8045	Core Model (I-III) 0.4391		
IV: $\pi_{vn}\pi_{m\theta} = \pi_{v\theta}\pi_{mn}$	0.0879				0.0356			
V: $\pi_{m\theta} = 0$	0.9790				0.9452			

^aThe sample without controls includes 257 cities, while the sample with controls includes 218 cities.

Table 2: Determinants of Skyscraper Heights, Land Values, Construction Input Prices, and Income:
Reduced Form Regression Results and Structural Restrictions Tests with Population Rank

Dependent Variables	Skyscr. height h^j (1)	Land value r^j (2)	Const. input v^j (3)	H-hold income m^j (4)	Metro pop. n^j (5)	Skyscr. height h^j (6)	Land value r^j (7)	Const. input v^j (8)	H-hold income m^j (9)	Metro pop. n^j (10)
<i>Panel A: Reduced-Form Parameter Estimates^a</i>										
Pop. rank, inverse (p^j)	0.361 (0.038)	1.006 (0.074)	0.050 (0.009)	0.087 (0.008)	0.959 (0.044)	0.337 (0.038)	0.977 (0.059)	0.050 (0.009)	0.081 (0.008)	0.919 (0.039)
Arc of expansion (log radians θ^j)	-0.176 (0.227)	-0.755 (0.455)	-0.073 (0.044)	0.016 (0.045)	0.085 (0.118)	-0.089 (0.151)	-0.854 (0.471)	-0.084 (0.045)	0.019 (0.034)	0.146 (0.088)
Regulatory Index (Wharton)						-0.098 (0.041)	0.280 (0.095)	-0.002 (0.012)	0.038 (0.017)	0.100 (0.033)
Building Age, years						-0.478 (0.157)	0.355 (0.279)	0.005 (0.034)	0.012 (0.038)	-0.165 (0.056)
Distance to bedrock (cms)						-0.018 (0.071)	0.219 (0.149)	0.037 (0.021)	0.053 (0.028)	0.040 (0.034)
R^2	0.738	0.841	0.538	0.698	0.967	0.788	0.851	0.554	0.731	0.973
<i>Panel B: Tests of Structural Restrictions^b</i>										
I: $\pi_{hp}(\pi_{r\theta} - \pi_{v\theta}) = \pi_{h\theta}(\pi_{rp} - \pi_{vp})$	0.8181			0.0303		0.4447		0.0008		
II: $\pi_{rp} = \pi_{np} + \pi_{mp}$	0.5429	Main Model + Zipf (I-III, V & VI-VII)				0.7125	Main Model + Zipf (I-III, V & VI-VII)			
III: $\pi_{r\theta} = -1 + \pi_{n\theta} + \pi_{m\theta}$	0.7546		0.5089			0.9676		0.0383		
IV: $\pi_{vp}\pi_{m\theta} = \pi_{v\theta}\pi_{mp}$	0.1372	Main Model (I-III & V)				0.0443	Main Model (I-III & V)			
V: $\pi_{mp}\pi_{N\theta} = \pi_{np}\pi_{m\theta}$	0.8548		0.6320			0.8605		0.6915		
VI: $\pi_{np} = 1$	0.3496	Core Model (I-III)				0.0349	Core Model (I-III)			
VII $\pi_{n\theta} = 0$	0.4716		0.4847			0.0925		0.7539		

^aThe sample without controls includes 257 cities, while the sample with controls includes 218 cities.

^bAdditionally, we performed a joint test of restrictions I-III and V, yielding p-values of 0.6320 for the results without controls and 0.69156 for the results with controls.

The building height specifications in columns 1 and 5 add variables to the specification first discussed in Figure 1, panel A. The coefficients on population and arc each have their predicted sign: the positive coefficient for population is rather precise and close to 0.4, while the negative coefficient for arc is negative, smaller, and imprecise. The estimates are consistent with each other as restriction I, seen in Panel B, passes at a p-value of 65 and 29 percent without and with controls. In column 5, both the regulatory index and building age are associated with shorter buildings, as expected, but the partial effect of distance to bedrock is insignificant, albeit negative.¹³ Overall the fit of the equation is strong, with an R-squared of 0.78 from land values alone, as seen Figure 1, rising to 0.83 with the additional variables.

The key land value results in columns 2 and 5 build on Figure 1, panel C. Moreover, the coefficient on population is close to one, and within range of one plus any standard value of the agglomeration parameter, γ , i.e., less than 10 percent. Remarkably, the coefficient on the arc of expansion is very close to minus one, as predicted, although it is imprecise. As seen in Panel B, the estimates here are consistent with restrictions II and III, meaning central land values are proportional to aggregate income and are inversely proportional to land availability. Bedrock and building age are statistically insignificant as controls, while the regulatory index is associated with higher central land values. As mentioned earlier, for a given population, this is unsurprising, as lower densities should cause a city to extend out further, much like a lower arc of expansion. The fit of this equation is the highest of all four, with an R-squared of over 0.8.

Columns 3 and 7 show the results for construction input prices. The coefficient on population is positive and small, as predicted.¹⁴ However, the coefficient on arc is significantly negative, meaning that construction costs are higher in more constrained cities, conditional on population. Not surprisingly, the corresponding structural restriction, IV, is strained with a p-value of 8 and 4 percent, without and with controls. Thus, we might expect low-arc cities to build taller because of higher land values, but shorter due to higher construction costs. None of the control variables are highly significant in this equation, although distance to bedrock is marginally positive. The R-squared, while high for a cross-sectional fit, is the lowest of all five equations.

Income is modeled in columns 4 and 8, the estimated coefficient on population is positive, precise, and relatively small, as expected. The estimated coefficient on arc of expansion is very close to zero, albeit imprecise, easily passing restriction V. The overall fit of the model is decently strong, with an R-squared of almost 0.7. There also appears to be a positive partial correlation of regulation with local income, conditional on population.

¹³We checked for a non-linear relationship with distance to bedrock but found nothing significant.

¹⁴It is less than the coefficient for income in columns 4 and 8.

Each of the individual structural restrictions fail to reject at a significance level of 20, except for restriction IV. This one violation appears to cause the joint test of all of the restrictions in the Full model to fail, with a p-value close to one percent. Without this restriction in the Main model, the joint significance passes at levels of 20 percent. The same holds for Core model, which ignores restriction V.

The results seen in Table 2 substitute in population rank for population, modeling Zipf’s Law in columns 5 and 10 in reduced-form equation (8). Unsurprisingly, the fit of this fifth equation is rather strong, with an R-squared of almost 0.97. The core Zipf restriction of a coefficient of one on population rank works better without controls. Both the regulatory index and building age significantly capture small deviations from this law, so that restriction VI fails at a significance level of 5 percent. Restriction VII also strains with the addition of the controls.

Nevertheless, for the most part, these other estimates do not differ tremendously from Table 1 when inverse population rank is substituted for population. Most importantly, land values rise with population at an elasticity of $1.049 = 1.006/0.959$, just slightly and insignificantly over one, hinting at agglomeration effects.

Without controls, all of the individual restrictions pass at significance levels of 10 percent and even above. Even the joint test of all seven restrictions, i.e., the Full model with Zipf’s Law, does rather well at passing at a significance level of 3 percent. Once restriction IV is dropped, the remainder of the model does quite well. With controls, the Zipf’s Law restrictions strain the model, which even at 1 percent in the Full Model, and even at 5 percent without restriction IV. Once the Zipf’s Law restrictions are allowed to loosen, the remainder of the model performs well again, passing at 50 percent significance. Thus population rank produces estimates consistent with the land value and building heights model, once imperfections in Zipf’s Law are accommodated.¹⁵]]

Table 3 presents the structural estimates of the parameters using Generalized Method of Moments (GMM). It presents estimates using two sets of restrictions. The columns with “Tight” restrictions (1) and (3) uses all five restrictions, while the columns with “Loose” restrictions allow the arc of expansion to impact construction costs and building heights.

The elasticity of substitution in building height is near 0.35 in almost all of the specifications. This is close to comparable estimates based on land-value variation in Chicago estimated by Ahlfeldt and McMillen (2018). Taking their estimates to convert this into a true elasticity of substitution of capital and land produces values slightly above 0.5.

The implied agglomeration parameter, γ is around 9 percent. This is slightly on the high side

¹⁵Versions of tables 1 and 2 with additional specification tweaks—variable definition and additional controls—are presented in an online appendix available at XXXX. In general, the results are robust to these changes in specification.

Table 3: Structural Estimation Results Using Population Rank

Model Controls	Main		Main + Zipf's Law	
	No (1)	Yes (2)	No (3)	Yes (4)
Elasticity of substitution on bldg hgt (σ^*)	0.333 (0.034)	0.379 (0.045)	0.336 (0.035)	0.351 (0.037)
Const. cost share of labor (a)	0.504 (0.104)	0.640 (0.140)	0.472 (0.117)	0.628 (0.129)
Population elasticity of income (γ)	0.085 (0.008)	0.085 (0.009)	0.087 (0.008)	0.090 (0.009)
Overidentification test	0.091	0.553	0.223	0.884

relative to most recent estimates, but not implausible for this parameter.

The labor share in construction parameter a is in most specifications close to one half, which is not far from most accounting procedures.

6 Discussion and Conclusion

We demonstrate that tight the relationship between skyscraper height and land values is explained through land values by marrying two canonical models. The first is a monocentric city model that predicts that central land values increase proportionally with aggregate income, and fall proportionally with a city's arc of expansion. Remarkably, the data appear consistent with the model's exact one-to-one predictions despite its radical simplicity.

The model relating land values to building heights results produces an estimate consistent with the literature. More interestingly, the two models combined justify a testable instrumental variable model. If the exclusion restrictions hold, a metro area's arc of expansion should lower building heights to the same degree that aggregate income raises them. In other words, skyscrapers are taller in places where geography makes it hard for cities to sprawl, and this operates through higher land values. While this prediction is not completely perfect, the core of the model survives a battery of tests, with greater reliability than Zipf's Law alone.

Furthermore, the model provides a tractable and plausible system of equation's that tracks several important economic urban economic phenomena together. It highlights visible features of

cities, taking together several complex phenomena, including urban agglomeration economies in a transparent framework. Thus, we hope it provides a good starting point for teaching and learning as well as future research.

For instance, fixing population, a city centered on a straight coast (like Chicago), should have double the land values of a city on an open plain. This echoes some of the more recent work on city shape such as [Harari \(2020b\)](#). Using a less known measure of this arc by Stephen Malpezzi, the rather tight prediction finds support in the data.¹⁶

¹⁶Using The measure is slightly different than that measured by [Saiz \(2010\)](#), which is the fraction of land over water or with a slope greater than 15 percent within 50 kilometers. Ours is more focused on the city center.

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A Appendix

A.1 Economic Model

A.1.1 Land Value Determination

This proof is based on [Combes et al. \(2012\)](#), an earlier version of [Combes et al. \(2016\)](#). Based on mobility within city, the equilibrium price of housing at distance x from the center falls in proportion to the increase in commuting costs, $C(x)$

$$Q(z) = Q(0) \left[\frac{C(0)}{C(x)} \right]^{\frac{1}{s}}$$

Housing prices relate to land price via the following unit cost function, where ϕ is the cost share of land in housing:

$$Q(x) = \frac{[R(x)]^\phi V^{1-\phi}}{A_Y \phi^\phi (1-\phi)^{1-\phi}}$$

Total housing demand at distance x is

$$N(x)Y(x) = N(x)s \frac{M}{Q(x)} = N(x) \frac{sM}{Q(0)} \left[\frac{C(x)}{C(0)} \right]^{\frac{1}{s}} = N(x) \frac{sMA_Y \phi^\phi (1-\phi)^{1-\phi}}{[R(0)]^\phi V^{1-\phi}} \left[\frac{C(x)}{C(0)} \right]^{\frac{1}{s}}$$

Note that the price gradient does not depend heavily on the assumption of constant income and cost shares: they are valid first-order approximations. The demand and supply quantities depend on the elasticity of substitution even in a first-order approximation, making them much more dependent on the Cobb-Douglas assumption.

Land values are determined by the derived demand for housing

$$R(z) = [Q(x)A_Y]^{\frac{1}{\phi}} \phi \left(\frac{1-\phi}{V} \right)^{\frac{1-\phi}{\phi}} = \underbrace{[Q(0)A_Y]^{\frac{1}{\phi}} \phi \left(\frac{1-\phi}{V} \right)^{\frac{1-\phi}{\phi}}}_{R(0)} \left[\frac{C(x)}{C(0)} \right]^{-\frac{1}{s\phi}}$$

Total housing supply at distance x , and substituting in for land values

$$Y^{TOT}(x) = x\Theta A_Y \left[\frac{R(x)}{V} \right]^{1-\phi} \left(\frac{1-\phi}{\phi} \right)^{1-\phi} = x\Theta A_Y \left[\frac{C(0)}{x} \right]^{1-\phi} \left(\frac{1-\phi}{\phi} \right)^{1-\phi} \left[\frac{C(x)}{C(0)} \right]^{-\frac{1-\phi}{s\phi}}$$

Taking the ratio of supply to demand to solve for the population at distance z

$$N(x) = \frac{Y^{TOT}(x)}{Y(x)} = R(0) \frac{\Theta x}{sM} \left[\frac{C(x)}{C(0)} \right]^{-\frac{1}{s\phi}}$$

The population at any distance is proportional to the central land value

$$N^{TOT} = \int_0^\infty N(z) dx = R(0) \frac{\Theta}{sM} \underbrace{\int_0^\infty \left[\frac{C(x)}{C(0)} \right]^{-\frac{1}{s\phi}} x \cdot dx}_{\equiv F}$$

Remarkably the commuting cost function may be taken out if commuting cost structures are similar across cities.¹⁷ Solving for $R(0)$ and taking the differential provides the following cost-based equation

$$d \ln R(0) = d \ln N - d \ln \Theta + d \ln M - d \ln F \quad (\text{A.1})$$

where using our notation for logarithms produces (2).¹⁸

A.1.2 Building Heights

Neither building height function nor floorspace is equivalent to capital. Rather, assume that capital investment rises proportionally with floor space F , but increases with building height as governed by the elasticity ζ

$$d \ln K = d \ln F + \zeta d \ln H \quad (\text{A.2})$$

A building has a certain footprint T , however the land required may rise with height, possibly from zoning codes

$$d \ln L = d \ln T + \xi d \ln H = d \ln F + (\xi - 1) d \ln H \quad (\text{A.3})$$

where the second equation uses the identity that height is merely floorspace divided by the footprint, $H = F/T \rightarrow T = F/H$ Thus we have that floorspace drops out capital to land ratio is

$$d \ln K - d \ln L = (1 + \zeta - \xi) d \ln H$$

We assume that skyscrapers compete for land with central residences. Unlike the latter, allow building of skyscrapers to follow a more flexible technology, with the elasticity of substitution between capital and land given by σ

$$d \ln K - d \ln L = \sigma(d \ln R(0) - d \ln V + d \ln B) \quad (\text{A.4})$$

and $d \ln B$ captures any differences due to factor bias. $\ln v$ is a construction cost index.

¹⁷If there is an outside option

¹⁸We approximate s constant across cities.

We then have

$$d \ln H = \underbrace{\frac{\sigma}{1 + \zeta - \xi}}_{\sigma^*} (d \ln R(0) - d \ln V + d \ln B) \quad (\text{A.5})$$

This provides equation (1). As discussed in the footnote, although We lack the data to estimate σ directly, we may compare it with other attempts in the literature, namely from [Ahlfeldt and McMillen \(2018\)](#).

A.2 Full Econometric Model with Zipf's Law

A.2.1 Structural Equations

The structural equations $\beta \mathbf{Y} + \Gamma \mathbf{Z} = \mathbf{U}$ are given as

$$\begin{bmatrix} 1 & -\sigma^* & \sigma^* & 0 & 0 \\ 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & -a & 0 \\ 0 & 0 & 0 & 1 & -\gamma \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} h \\ r(0) \\ v \\ m \\ N \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} p \\ \theta \end{bmatrix} = \begin{bmatrix} b \\ f \\ a_v \\ a_m \\ e_p \end{bmatrix}$$

Controls may put in the residual term $\mathbf{U}$ without loss of generality.

A.2.2 Reduced Form

$\mathbf{Y} = \Pi \mathbf{Z} + \mathbf{V}$, where $\Pi = -\beta^{-1}\Gamma$, which means that the reduced form coefficient matrix is

$$\begin{bmatrix} \pi_{hp} & \pi_{h\theta} \\ \pi_{rp} & \pi_{r\theta} \\ \pi_{vp} & \pi_{v\theta} \\ \pi_{mp} & \pi_{m\theta} \\ \pi_{np} & \pi_{n\theta} \end{bmatrix} = \begin{bmatrix} -1 & -\sigma^* & \sigma^* & -\sigma^*(1-a) & -\sigma^*(1+\gamma-a\gamma) \\ 0 & -1 & 0 & -1 & -1-\gamma \\ 0 & 0 & -1 & -a & -a\gamma \\ 0 & 0 & 0 & -1 & -\gamma \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ -1 & 0 \end{bmatrix}$$

$$\dots = \begin{bmatrix} \sigma^*(1+\gamma-a\gamma) & -\sigma^* \\ 1+\gamma & -1 \\ a\gamma & 0 \\ \gamma & 0 \\ 1 & 0 \end{bmatrix}$$

Here there are 7 restrictions and 3 free parameters. Written independently, these restrictions are

- I $\pi_{hp}(\pi_{r\theta} - \pi_{v\theta}) = \pi_{h\theta}(\pi_{rp} - \pi_{vp})$. This equivalent to the restriction that each instrument produces the same elasticity of substitution. $\sigma^* = \frac{\pi_{hp}}{\pi_{rp} - \pi_{vp}} = \frac{\pi_{h\theta}}{\pi_{r\theta} - \pi_{v\theta}}$.
- II $\pi_{rp} = \pi_{np} + \pi_{mp}$ central land values should rise proportionally with population save for an adjustment due to agglomeration raising income.

- III $\pi_{r\theta} = -1 + \pi_{N\theta} + \pi_{m\theta}$ central land values should fall proportionally with the arc of expansion, save for any effect through income or population (which other restrictions rule out).
- IV $\pi_{vp}\pi_{m\theta} = \pi_{v\theta}\pi_{mp}$ This checks that differences in construction input prices are due to wages alone. This may be written in terms of the cost parameter $a = \frac{\pi_{vp}}{\pi_{mp}} = \frac{\pi_{v\theta}}{\pi_{m\theta}}$. The arc of expansion cannot independently affect construction input prices.
- V $\pi_{mp}\pi_{n\theta} = \pi_{np}\pi_{m\theta}$. This rules out any direct relationship between the arc of expansion and income. This may be rewritten as $\gamma = \frac{\pi_{mp}}{\pi_{np}} = \frac{\pi_{m\theta}}{\pi_{n\theta}}$
- VI $\pi_{np} = 1$ This is the prediction of Zipf's Law.
- VII $\pi_{n\theta} = 0$ Deviations from Zipf's Law are not predicted by θ .

Online Appendix - Not for Publication

B Data

B.1 Building heights

We use data on building heights from skyscraperpage.com. An important part of our approach is having a measure of height consistent with the floor area, that is why we measure heights from the ground to the roof, leaving aside ornamental features as spires and antennas. For part of the sample our source only provided the floor count but not height in feet. For those building we uses the predicted height bases on the floor count. Figure (A.1) shows the relation between height and the number of floors, our estimate is that the average height of a floor is 11.4 feet.

Table A.1: Summary Statistics

	Mean	Std.Dev.	Obs
Height top 10 buildings	220.19	169.14	257
Land values (millions)	2.06	7.06	257
Construction cost	89.93	9.67	257
Household income	59680.85	9069.49	257
Population (millions)	0.84	2.09	257
Arc of expansion (percentage)	0.91	0.19	257
Wharton city index	-0.24	0.90	222
Age top 10 high-rises by 2020	52.58	21.25	256
Distance to Bedrock	2062.51	1217.97	254
Observations	257		

B.2 Arc of Expansion

Data on the arc of expansion from Stephen Malpezzi was complemented with our own calculations using Google maps. The method is simple, for cities that have shores with a large bodies of water (oceans or Great Lakes), we draw a circle with a radius no greater than 50 miles. Drawing a triangle as described in Figure (A.2), we calculate the degrees of expansion lost to the body of water.

Figure A.1: Building Height and the Number of Floors

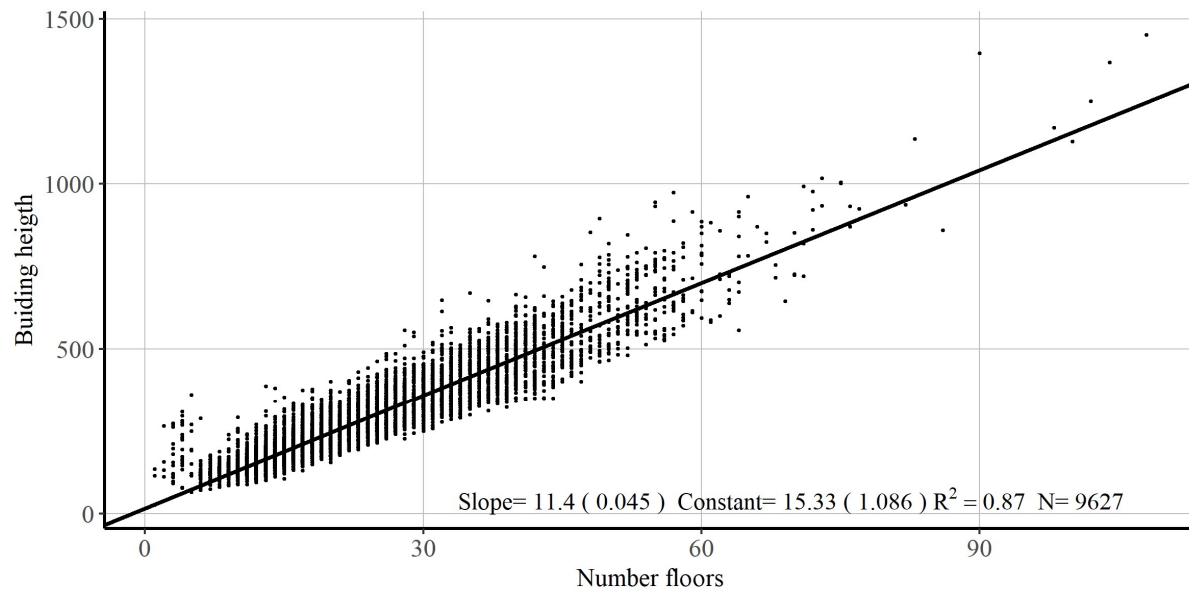
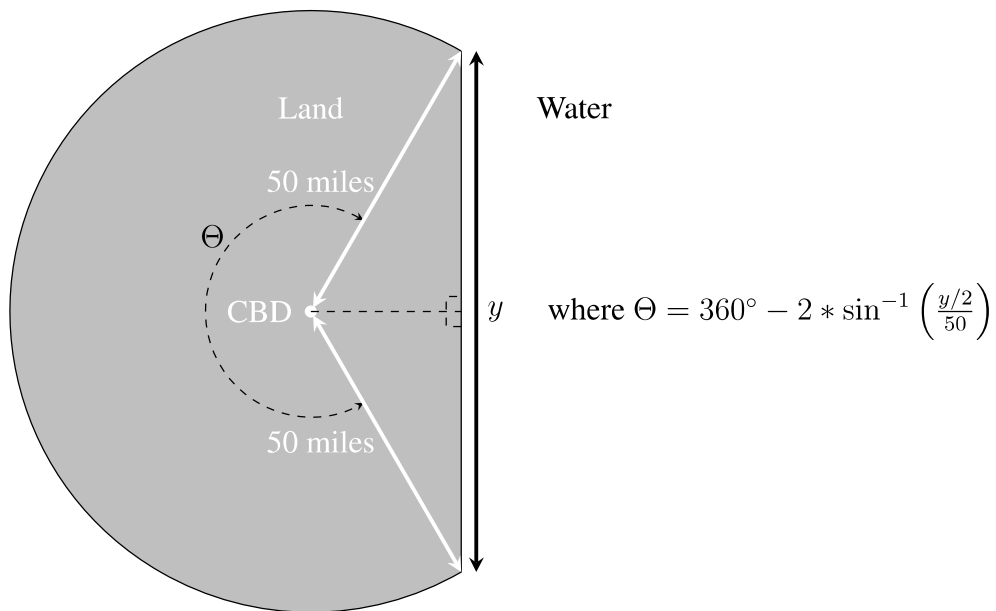


Figure A.2: Measure Arc of Expansion



C Regression Robustness Checks

To assess the robustness of our main findings, we test several alternative specifications for the results presented in Tables 1 and 2. We organize these robustness checks into four categories, described below.

C.1 Alternative Definition of Skyscraper Heights

The main estimates measure building height using the ten tallest buildings in each city. We chose this option for a number of reasons. First, to increase the number of cities we could use, as the heights of shorter buildings are often unavailable. Second, because the tallest buildings visually seem to best define cities' skylines. Third, by using taller buildings, we could work around many buildings being too short due to developers holding back on height because of local land-use politics, or to preserve option value for building higher. Fourth, as these buildings might be considered to be the most irrationally high, working against our hypothesis. To address the robustness of our results we adopt an alternative definition based on the 25 tallest buildings in each city. Table A.2 shows that our findings are robust to this alternative definition, as none of the statistically significant estimates exhibit meaningful changes.

C.2 Land Values Without Shrinkage

The land values provided by Albouy et al. (2018) rely a shrinkage technique, using empirical Bayes methods that allow us to infer the value of land even in cities with few or even no observations, based on their area. The direction of shrinkage in the coefficients of the land-value gradient function depends on the urban area of the city. The relationship implied by this approach differs from the one in the present paper, since we do not impose a specific functional relationship between central land values, population, and the arc of expansion. As a result, the shrinkage is not imposed toward the empirical relationship documented in the main analysis or toward the relationship predicted by our theoretical framework. Even in this case, we believe it is worthwhile to examine the sensitivity of our results to this modeling choice.

To examine this possibility, we re-estimate our specifications using an alternative measure of land values that does not rely on shrinkage methods. Specifically, we use the average land value within one mile of the city center. Table ?? shows that the coefficient on population in the land-value equation remains close to one. The point estimate for the coefficient on the arc of expansion also moves closer to one, although its standard error is large enough to encompass the corresponding estimate from the main specification. The direction and magnitude of the remaining coefficients are consistent with those in the main specification.

Overall, we find that the empirical results obtained using unshrunk estimates are close to those from our main specification. We therefore present results based on the shrunk estimates, which allow us to include a larger set of cities in the analysis.

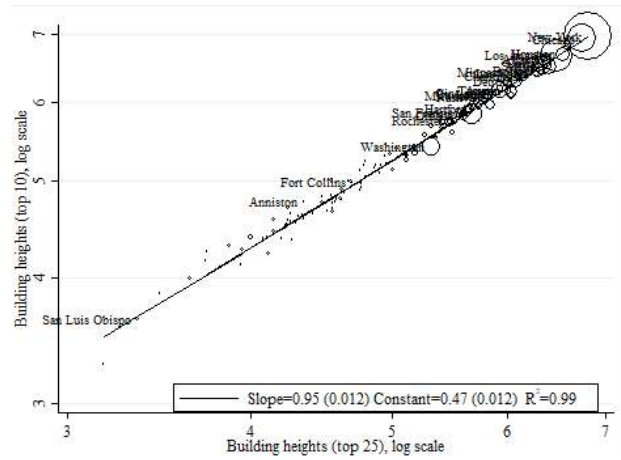


Figure A.3: Enter Caption

C.3 Measure of Land Availability

To address potential bias arising from the definition of land availability or from measurement error in the construction of this variable, we employ the measure used in [Saiz \(2010\)](#). This measures the percentage of land within a 50 km radius that is not covered by water or which has a slope over 15 percent. The results in Table A.3 confirm that our findings are robust to this change in variable definition, as none of the statistically significant estimates exhibit meaningful changes. Nevertheless, it is worth noting that none of the joint tests for the different model definitions—Full, Main, and Core—passed, even though none of the individual restrictions are rejected. We suspect this may be related to the statistically non-significant yet counterintuitive positive effect of land availability on building height that is estimated when using the measure from [Saiz \(2010\)](#).

C.4 Additional Controls

To address potential bias due to omitted variables, we include several additional controls: the square of the distance to bedrock, the share of the labor force with a bachelor's degree or higher, and the share of the labor force employed in business, finance, and legal occupations. The results presented in Table A.4 confirm that our findings are robust to the inclusion of these additional controls, as none of the statistically significant estimates exhibit meaningful changes.

Table A.2: Determinants of Skyscraper Heights and Land Values Using Log Population
Top 25 Tallest Buildings Estimates, Reduced Form Regression Results

Outcome	Skyscr. height h^j (1)	Land value r^j (2)	Const cost v^j (3)	Hhold income m^j (4)	Skyscr. height h^j (5)	Land value r^j (6)	Const cost v^j (7)	Hhold income m^j (8)
<i>Panel A: Reduced-Form Regression Results</i>								
Population n^j	0.389 (0.038)	1.043 (0.085)	0.050 (0.012)	0.089 (0.010)	0.401 (0.032)	1.026 (0.087)	0.053 (0.012)	0.084 (0.010)
Arc of expan, (log radians θ^j)	-0.233 (0.218)	-0.957 (0.560)	-0.104 (0.055)	0.016 (0.050)	-0.151 (0.173)	-1.096 (0.567)	-0.132 (0.054)	0.004 (0.042)
Wharton city index					-0.136 (0.042)	0.208 (0.119)	-0.009 (0.013)	0.032 (0.018)
Log age top 25 high-rises					-0.393 (0.124)	0.451 (0.367)	0.064 (0.037)	0.045 (0.046)
Distance to Bedrock					-0.027 (0.096)	0.226 (0.207)	0.033 (0.026)	0.058 (0.035)
R^2	0.742	0.775	0.474	0.630	0.788	0.782	0.526	0.664
# of metros	139	139	139	139	122	122	122	122
<i>Panel B: Tests of Structural Restrictions</i>								
I: $\pi_{hn}(\pi_{r\theta} - \pi_{v\theta}) = \pi_{h\theta}(\pi_{rn} - \pi_{vn})$	0.7896	Full Model (I-V) 0.0667			0.4539	Full Model(I-V) 0.0082		
II: $\pi_{rn} = 1 + \pi_{mn}$	0.5690	Main Model (I-III & V) 0.7756			0.4717	Main Model (I-III & V) 0.6902		
III: $\pi_{r\theta} = -1 + \pi_{m\theta}$	0.9588	Core Model (I-III) 0.7789			0.8493	Core Model (I-III) 0.6752		
IV: $\pi_{vn}\pi_{m\theta} = \pi_{v\theta}\pi_{mn}$	0.0718				0.0216			
V: $\pi_{m\theta} = 0$	0.7419				0.9198			

Table A.3: Determinants of Skyscraper Heights and Land Values Using Log Population
Saiz's Land Availability Measure, Reduced Form Regression Results

Outcome	Skyscr. height h^j (1)	Land value r^j (2)	Const cost v^j (3)	Hhold income m^j (4)	Skyscr. height h^j (5)	Land value r^j (6)	Const cost v^j (7)	Hhold income m^j (8)
<i>Panel A: Reduced-Form Regression Results</i>								
Log population	0.410 (0.029)	1.065 (0.051)	0.051 (0.012)	0.086 (0.007)	0.396 (0.031)	1.096 (0.071)	0.056 (0.012)	0.084 (0.009)
Saiz's availability	0.113 (0.201)	-0.577 (0.428)	-0.083 (0.038)	-0.041 (0.056)	0.115 (0.105)	-0.611 (0.407)	-0.111 (0.039)	-0.037 (0.043)
Wharton city index					-0.133 (0.043)	0.121 (0.137)	-0.018 (0.013)	0.025 (0.015)
Age of bldgs, years					-0.438 (0.172)	0.470 (0.279)	0.020 (0.030)	0.036 (0.044)
Distance to Bedrock					-0.041 (0.073)	0.163 (0.167)	0.032 (0.019)	0.049 (0.027)
R^2	0.782	0.811	0.500	0.704	0.837	0.812	0.540	0.718
# of metros	255	255	255	255	216	216	216	216
<i>Panel B: Tests of Structural Restrictions</i>								
I: $\pi_{hn}(\pi_{r\theta} - \pi_{v\theta}) = \pi_{h\theta}(\pi_{rn} - \pi_{vn})$	0.3591	Full Model (I-V) 0.0000			0.1440	Full Model(I-V) 0.0000		
II: $\pi_{rn} = 1 + \pi_{mn}$	0.6619	Main Model (I-III & V)			0.8581	Main Model (I-III & V)		
III: $\pi_{r\theta} = -1 + \pi_{m\theta}$	0.2403	0.0022			0.2701	0.0002		
IV: $\pi_{vn}\pi_{m\theta} = \pi_{v\theta}\pi_{mn}$	0.2014	Core Model (I-III)			0.0495	Core Model (I-III)		
V: $\pi_{m\theta} = 0$	0.4619	0.0051			0.3768	0.0002		

Table A.4: Determinants of Skyscraper Heights and Land Values Using Log Population
Additional Controls, Reduced Form Regression Results

Outcome	Skyscr. height h^j (1)	Land value r^j (2)	Const cost v^j (3)	Hhold income m^j (4)	Skyscr. height h^j (5)	Land value r^j (6)	Const cost v^j (7)	Hhold income m^j (8)
<i>Panel A: Reduced-Form Regression Results</i>								
Log population	0.385 (0.036)	1.002 (0.071)	0.046 (0.010)	0.088 (0.009)	0.345 (0.030)	0.934 (0.103)	0.048 (0.011)	0.057 (0.006)
Arc of expan, (log radians θ^j)	-0.186 (0.207)	-0.966 (0.491)	-0.094 (0.049)	0.001 (0.045)	-0.109 (0.133)	-1.100 (0.526)	-0.094 (0.049)	0.010 (0.025)
Wharton city index					-0.109 (0.035)	0.172 (0.106)	-0.002 (0.012)	0.026 (0.007)
Age of bldgs, years					-0.324 (0.127)	0.383 (0.329)	-0.010 (0.034)	-0.002 (0.021)
Distance to Bedrock					0.908 (0.840)	0.681 (2.507)	0.469 (0.283)	0.399 (0.214)
Square Distance to Bedrock					-0.063 (0.056)	-0.038 (0.167)	-0.029 (0.020)	-0.025 (0.014)
Prop. Labor force in business, finance, and legal					0.798 (0.214)	-0.362 (0.617)	-0.030 (0.073)	-0.045 (0.041)
Prop. Labor force with BA or higher					-0.565 (0.327)	1.425 (0.541)	0.029 (0.082)	0.374 (0.050)
R^2	0.783	0.816	0.482	0.689	0.851	0.828	0.511	0.853
# of metros	257	257	257	257	195	195	195	195
<i>Panel B: Tests of Structural Restrictions</i>								
I: $\pi_{hn}(\pi_{r\theta} - \pi_{v\theta}) = \pi_{H\theta}(\pi_{rn} - \pi_{vn})$	0.6484	Full Model (I-V) 0.0103			0.0093	Full Model(I-V) 0.0004		
II: $\pi_{rn} = 1 + \pi_{mn}$	0.2086	Main Model (I-III & V) 0.3676			0.4051	Main Model (I-III & V) 0.0002		
III: $\pi_{r\theta} = -1 + \pi_{m\theta}$	0.9431	Core Model (I-III) 0.2323			0.0080	Core Model (I-III) 0.0012		
IV: $\pi_{vn}\pi_{m\theta} = \pi_{v\theta}\pi_{mn}$	0.0879				0.1673			
V: $\pi_{m\theta} = 0$	0.9790				0.0000			